LALITHYA VANACHARLA

■ vanacharlalalithya@gmail.com | ■ +91 9059531778

■ www.linkedin.com/in/lalithya-vanacharla-49a0a5331/

PROFILE SUMMARY

As a B.Tech. IT student with strong programming skills in Python and Java, seeking an entry-level position in the IT sector where I can apply my coding expertise, problem-solving abilities, and passion for technology to develop efficient solutions and contribute to organizational growth while continuously enhancing my technical skills.

EDUCATION

B. Tech in Information Technology – Sagi Rama Krishnam Raju Engineering College, Bhimavaram (2022 – 2026), CGPA: 8.16
Intermediate (MPC) – Sasi Junior College for Girls, Velivennu (2020 – 2022), Percentage: 93.4%
Class 10 – Sasi English Medium High School, Velivennu (2020), Percentage: 80.6%

EXPERIENCE

Python Full Stack Development Intern – IIDT Blackbucks (05/2024 – 07/2024)

- Engineered and deployed 3+ web applications using Python (Django, Flask), HTML, CSS, JavaScript, reducing deployment cycle time by 25%.
- Designed and integrated RESTful APIs, improving data transfer efficiency by 30% and supporting 1,000+ API calls/day.
- Optimized MySQL/PostgreSQL databases, cutting query time by 40% on datasets exceeding 100,000+ records.
- Implemented secure authentication & CRUD modules, increasing system reliability by 20% and reducing security flaws.

Artificial Intelligence and ML Engineer Intern - IIDT Blackbucks (22/12/2025 - 21/04/2026)

- Enforced and trained 8+ machine learning models (Logistic Regression, SVM, Random Forest), achieving 92% accuracy and improving baseline performance by +27%
- Processed and cleaned 50,000+ rows of structured data, reducing missing/inconsistent values by 65% through automated preprocessing pipelines
- Engineered 20+ features, boosting model performance (F1-score) from 0.68 → 0.89 (+31%)
- Reduced model training time by 45% using optimized data pipelines and vectorized operations in NumPy

PROJECTS

Weather Prediction Application

• Developed a Python based ML model achieving 85%+ prediction accuracy for rainfall & temperature forecasts.
• Processed 10,000+ weather records, cutting prediction error rates by 20% through advanced feature engineering.
• Automated real-time weather updates via Open Weather Map API, reducing manual input by 100%.
• Created a TKinter interface, enabling 100+ end-users to access accurate forecasts daily.
Tools & Technologies: Python, Machine Learning, TKinter, Open Weather Map API, Matplotlib

Auto Scaling with Load Balancer on AWS-Online Movie Ticket Booking Platform

• Configured AWS Auto Scaling Groups & Load Balancer, handling 2x peak traffic spikes with 0% downtime.
• Reduced infrastructure costs by 25% through dynamic resource allocation.

• Enhanced application availability by 40% and fault tolerance by 35% by distributing workloads across multiple EC2 instances.

Tools & Technologies: AWS EC2, Auto Scaling, Load Balancer, Cloud Watch, Basic DevOps

A Hybrid Approach for Multi-Class Brain Tumor Classification Using Ensemble CNN and PSO Optimization

- Designed a hybrid ensemble framework combining 3 CNN architectures (custom CNN + pre-trained variants) to classify 4 tumor classes (glioma, meningioma, pituitary, no tumor)
- Instructed on ~7,500 MRI images (80:20 train-test split), achieving 95.4% test accuracy and +18% improvement over single CNN baseline
- Implemented weighted averaging ensemble, improving F1-score from 0.81 → 0.93 (+14.8%) and reducing class imbalance impact
- Performed data augmentation (6 techniques) expanding dataset variability by ~45%, reducing overfitting gap from 12% → 4%

Tools & Technologies: Python, TensorFlow / Keras, OpenCV, NumPy, Pandas, Matplotlib, CNN, PSO

SKILLS

Languages: Python, Java

Front end :HTML,CSS

Database: MySQL

Soft Skills: Communication, Problem Solving, Team work

CERTIFICATIONS

- **NPTEL**: Machine Learning and Data Science Certification
- **IIDT** Blackbucks: AWS Cloud Computing & DevOps Basics, Full Stack Development using Python, AI-ML Engineer
- **Be10x**: AI Tools and ChatGPT Workshop Certification